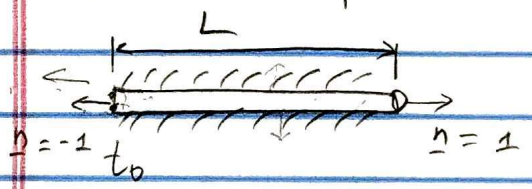
Prob. 6. (2 & T. Ch. 6. 2.3)

Write an initial-boundary value prob-m (IBVP) that must be solved to determine the history of temperature distribution in a thin [added to the pr-m from the text] cylindrical rod of length L with insulated cylindrical surface, given the initial temp-re distribution in the rod at some time, $t = t_0$, and the temperature at the two ends of the rod for all time, $t \geq t_0$.

Remark: The IBVP includes the governing eq-n, all boundary conditions, & initial condition. Be sure the domain is specified for each eq-n.

Sol-n:



$u = u(x,t) =$ temperature of the rod at time $t \geq t_0$
$x \in [0, L]$

(1) $u_t = u_{xx}$ on $\Omega = \{(x,t): 0 < x < L,\ t_0 < t < \infty\}$

$u(x, t_0) = f(x) \qquad 0 < x < L$
$u(0,t) = g(t) \qquad t \geq t_0$
$u(L,t) = h(t) \qquad t \geq t_0$

Pr. 8. (i) – D (iii) – A
(ii) – E (iv) – B